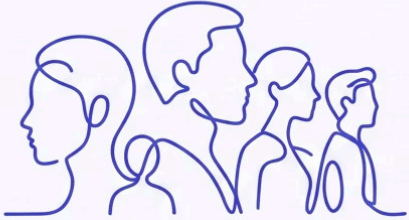




Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and product preferences from **3,900 transactions** to guide strategic decisions.

Dataset at a Glance



3,900 Customers
**Customer
Demographics**



18 Columns
**Purchase
Details**



**Shopping
Behavior**



Data Quality
37 missing values

What's Inside

- **3,900** purchase records
- **18** columns covering demographics, purchases, and behavior
- **37** missing values in Review Rating — imputed by category median



Data Preparation in Python

Clean & Standardize

Snake_case columns, median imputation for missing ratings

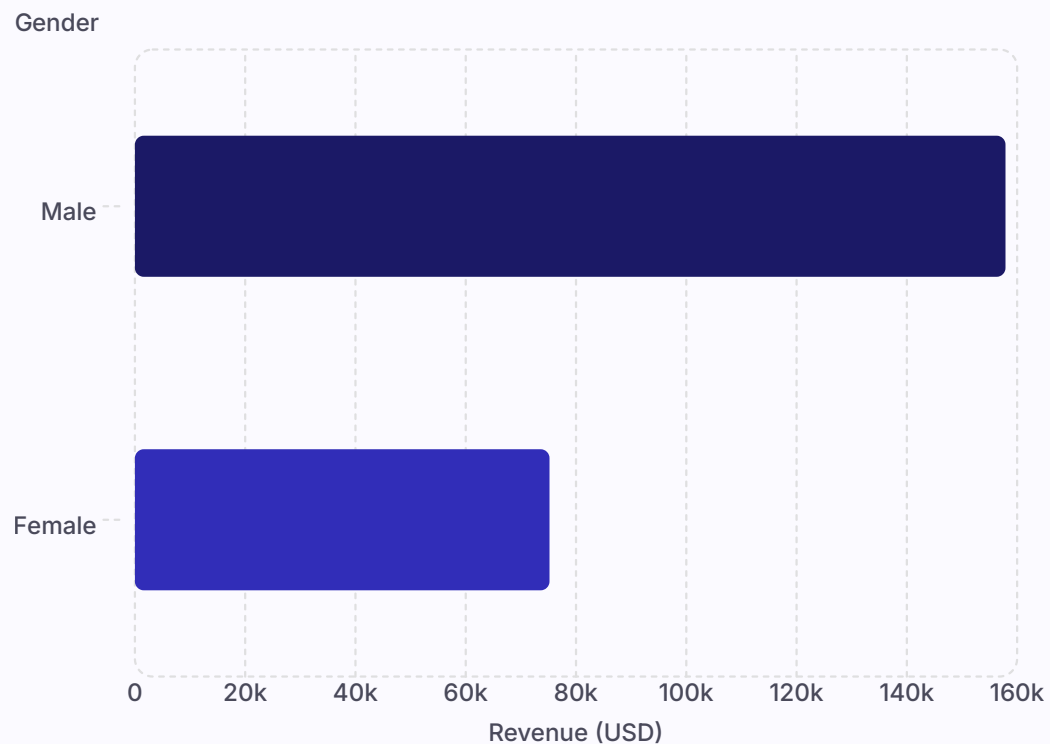
Feature Engineering

Created **age_group** and **purchase_frequency_days**

Database Load

Cleaned DataFrame loaded into **PostgreSQL** for SQL analysis

Revenue by Gender



Key Finding

Male customers generated **2.1× more revenue** than female customers.

Male: \$157,890 vs. Female: \$75,191

Top Products & Discount Patterns

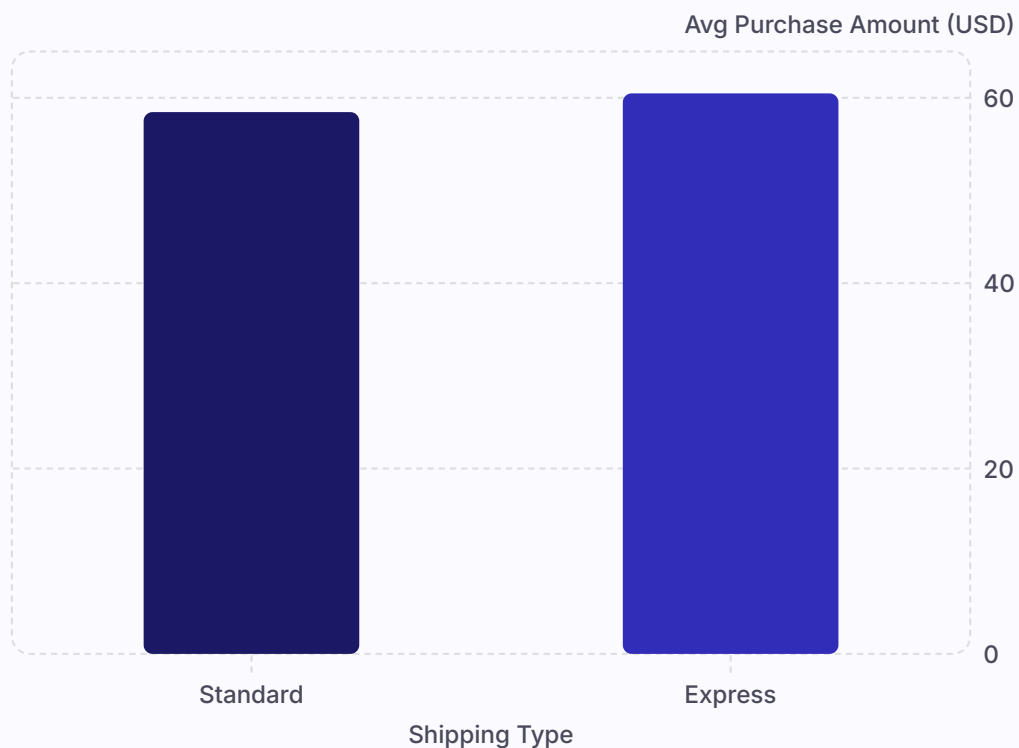
Top 5 Products by Review Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Shipping & Subscription Insights



Subscribers vs. Non-Subscribers

Subscribers (27%)

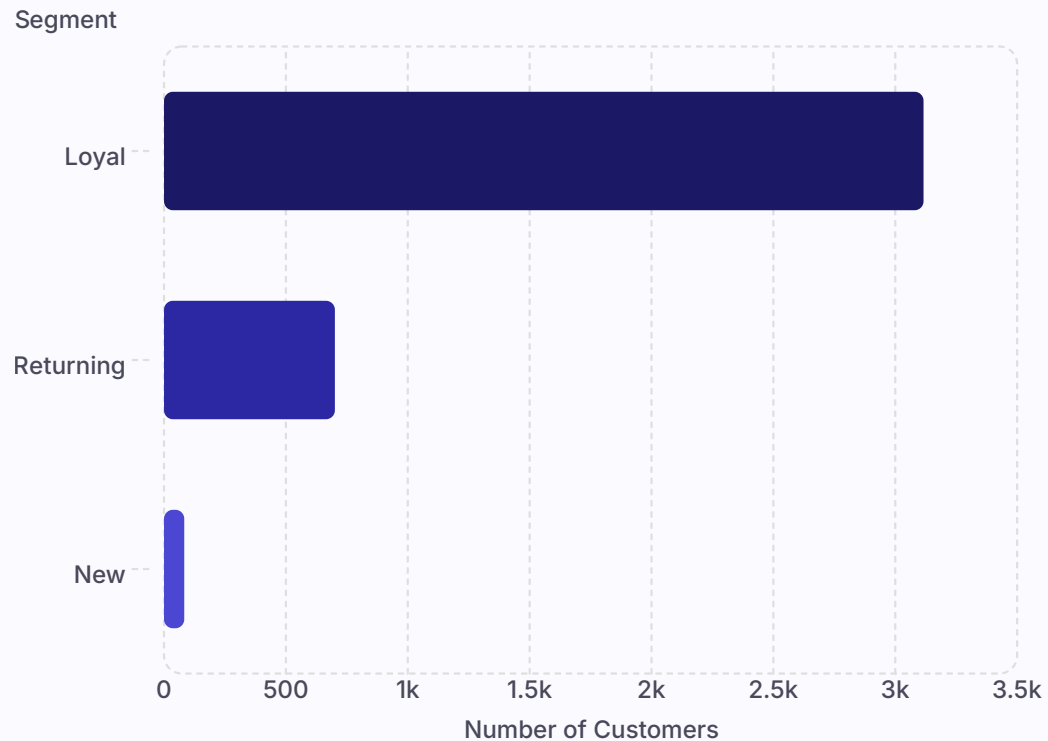
1,053 customers · Avg
\$59.49

Non-Subscribers (73%)

2,847 customers · Avg
\$59.87

Similar avg spend — subscription drives **loyalty, not basket size.**

Customer Segmentation



Segment Breakdown

Loyal — 80%

3,116 customers

Returning — 18%

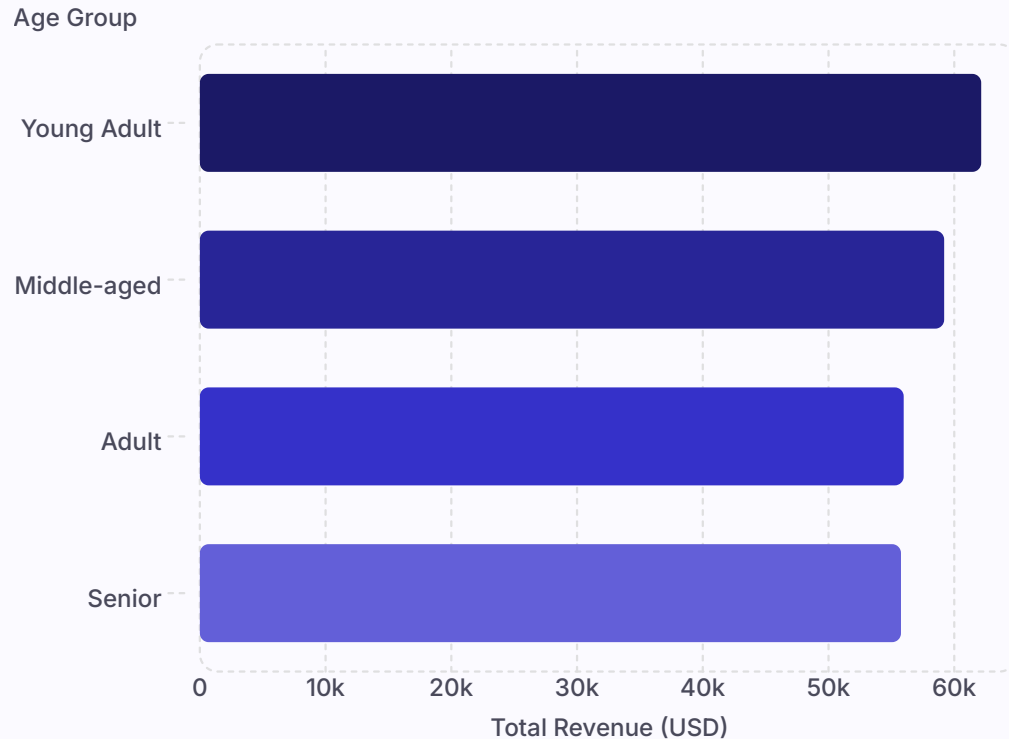
701 customers

New — 2%

83 customers

958 of 3,116 loyal customers are subscribers — strong repeat-buyer overlap.

Revenue by Age Group & Category



Top Products per Category

- **Accessories:** Jewelry, Sunglasses, Belt
- **Clothing:** Blouse, Pants, Shirt
- **Footwear:** Sandals, Shoes, Sneakers
- **Outerwear:** Jacket, Coat

Young Adults lead revenue; Clothing dominates all categories.

Power BI Dashboard



Key Metrics

3.9K customers · \$59.76 avg purchase · 3.75 avg rating



Interactive Filters

Subscription, Gender, Category, Shipping Type



Visualizations

Revenue & Sales by Category and Age Group



Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers → Loyal segment

Review Discount Policy

Balance sales lift with margin control

Product Positioning

Highlight top-rated & best-selling items

Targeted Marketing

Focus on high-revenue age groups & express-shipping users